

The VS Code Extension Development Cycle

What You Built in Step 1

Step 1 used `yo code` to generate a scaffold. You pressed F5 and saw “Hello World.” But you didn’t write any of that code — a tool did. Step 2 is where you take ownership.

The Two Files That Matter

A VS Code extension has two halves that must agree with each other:

1. `package.json` — The Declaration

```
"contributes": {
  "commands": [
    {
      "command": "hello-zygote.helloWorld",
      "title": "Hello World"
    }
  ]
}
```

This tells VS Code: *“I offer a command. Its internal ID is `hello-zygote.helloWorld`. Show it to the user as ‘Hello World’ in the Command Palette.”*

This is **registration only**. It does not define what happens when the command runs. Think of it as a menu item with no function attached yet.

2. `src/extension.ts` — The Implementation

```
const disposable = vscode.commands.registerCommand(
  'hello-zygote.helloWorld', // must match package.json exactly
  () => {
    vscode.window.showInformationMessage('Hello World from hello-zygote!');
  }
);
```

This tells VS Code: *“When someone triggers `hello-zygote.helloWorld`, run this function.”*

The Contract

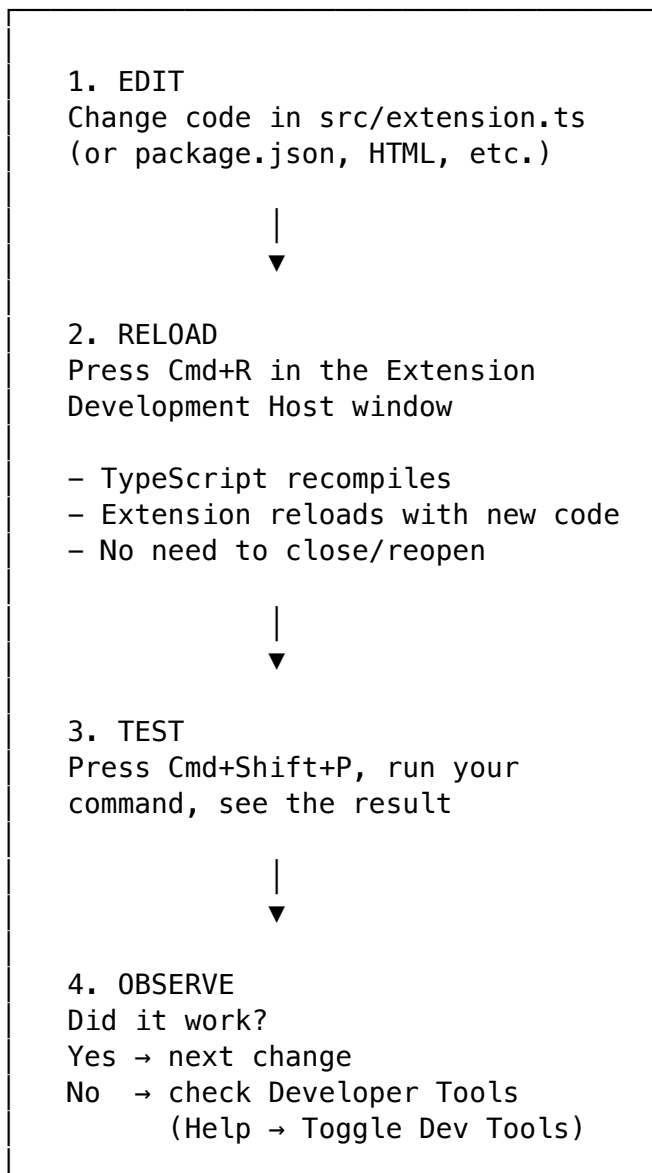
package.json	extension.ts	Result
Has command ID	Has matching <code>registerCommand</code>	Works
Has command ID	No matching <code>registerCommand</code>	Command appears but does nothing

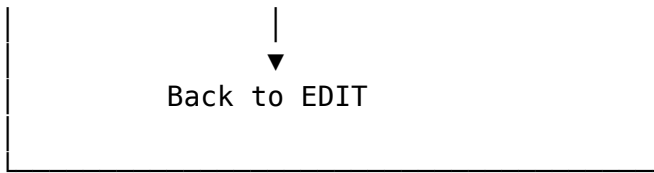
package.json	extension.ts	Result
No command ID	Has registerCommand	Command never appears in palette
IDs don't match	IDs don't match	Broken — silent failure

Key insight: The command ID string (`hello-zygote.helloWorld`) is the bridge between declaration and implementation. If these two files disagree on that string, the extension silently fails.

The Development Cycle

Every change you make to a VS Code extension follows this loop:





When to use Cmd+R vs. Fn+F5

Action	When to use
Cmd+R (reload)	You changed code inside <code>src/extension.ts</code> — logic, messages, behavior
Fn+F5 (full restart)	You changed <code>package.json</code> — new commands, activation events, metadata

Why the difference: Cmd+R reloads the extension code but keeps the same Extension Development Host. Changes to `package.json` require a full restart because VS Code reads that file only at launch.

What “Step 2” Asks You To Do

Exercise 1: Change the message

Open `src/extension.ts`, find this line:

```
vscode.window.showInformationMessage('Hello World from hello-zygote!');
```

Change the string to anything — your name, a joke, a question. Save the file.

Exercise 2: Reload

In the Extension Development Host window, press Cmd+R.

Exercise 3: Test

Press Cmd+Shift+P, type Hello World, press Enter.

Exercise 4: Verify

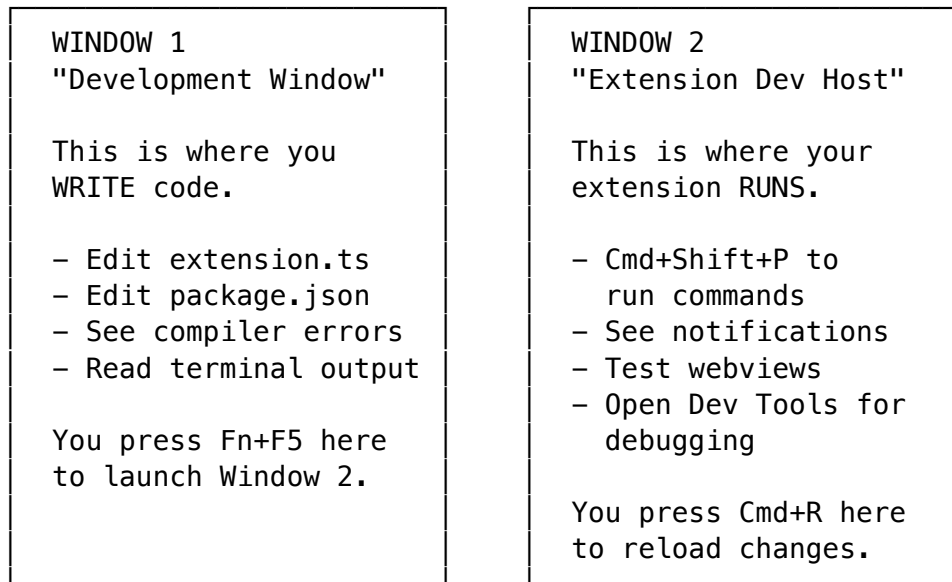
You should see your custom message in the bottom-right corner. If you do, you have proven:

- You know which file controls the behavior
- You know how to push a change into the running extension
- You know how to trigger and observe the result

This is the entire development cycle. Every feature in Zygote — webview, API calls, file reading — is built by repeating this loop.

The Two Windows

When developing a VS Code extension, you always have two VS Code windows open:



The activate() Function

```
export function activate(context: vscode.ExtensionContext) {  
    // Everything inside here runs ONCE when the extension first activates  
    // "Activates" = the first time a user triggers one of your commands  
}
```

VS Code does **not** run your extension at startup. It waits until the user actually needs it (lazy activation). This keeps VS Code fast even with many extensions installed.

context is your extension's lifetime manager. Anything you push into context.subscriptions gets automatically cleaned up when the extension deactivates. This prevents memory leaks.

Summary

Concept	What it means
package.json	Declares what your extension offers (commands, menus, settings)
src/extension.ts	Implements the behavior behind those declarations
Command ID	The string that connects declaration to implementation
activate()	Runs once when the extension is first needed

Concept	What it means
<code>context.subscriptions</code>	Automatic cleanup for event listeners and commands
Cmd+R	Reload extension code without restarting
Fn+F5	Full restart (needed after package.json changes)
Cmd+Shift+P	Open Command Palette to run commands
Dev Tools	Your debugger when something goes wrong

The development cycle is: **Edit** □ **Reload** □ **Test** □ **Observe** □ **Repeat**.

Master this loop. Everything else builds on top of it.